

Instagram User Analytics

SQL & Tableau Analytics Project

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Instagram User Analytics

Executive Summary

This project analyses user behaviour on an Instagram-like social media platform using SQL and Tableau. The objective was to evaluate user engagement patterns, identify opportunities for marketing intervention, assess content performance, and investigate indicators of potentially automated behaviour.

Analysis of 100 users, 257 photos, and 8,782 likes revealed three key findings. First, user engagement is highly uneven, with 26% of registered users never posting any content. Second, user acquisition appears strongest on Thursdays and Sundays, suggesting optimal timing for future marketing campaigns. Third, a subset of accounts demonstrated suspiciously high engagement behaviour by liking every photo on the platform, indicating potential bot activity.

The findings provide actionable recommendations for improving user retention, increasing content creation, and strengthening platform integrity.

1. Introduction

Social media platforms rely heavily on user engagement and content creation to sustain growth. Understanding how users interact with the platform enables marketing, product, and leadership teams to make informed decisions regarding customer acquisition, retention, and monetisation.

This project investigates Instagram user activity through SQL-based exploratory analysis and Tableau visualisation. The analysis addresses both operational marketing questions and strategic investor-focused metrics.

2. Methodology

The analysis was conducted using MySQL Workbench for data extraction and Tableau for visualisation.

The dataset consisted of seven relational tables representing users, photos, likes, comments, hashtags, and photo-tag relationships.

SQL queries were developed to answer specific business questions relating to:

- User loyalty
- User inactivity
- Content performance
- Hashtag popularity
- User acquisition trends
- User engagement
- Potential bot behaviour

Results were subsequently visualised through an interactive dashboard to facilitate business interpretation.

Key Performance Indicator	Value
Registered Users	100
Photos Uploaded	257
Total Likes	8,782
Average Posts per User	2.57
Inactive Users	26
Potential Bot Accounts	13

3. Marketing Insights and User Behaviour Analysis

3.1 Loyal User Identification

Finding

The analysis identified the five earliest registered users on the platform, representing Instagram's longest-tenured user segment.

Business Implication

Long-tenured users often demonstrate stronger platform loyalty and engagement. These users can serve as valuable candidates for beta testing, referral programmes, loyalty rewards, and community-building initiatives.

Recommendation

Develop targeted loyalty campaigns to recognise and retain early adopters while leveraging their experience to support future product development and user feedback initiatives.

3.2 Inactive User Analysis

Finding

Twenty-six users were identified as having never uploaded a photo, representing 26% of the platform's total user base.

Business Implication

This suggests that user acquisition efforts are successfully driving registrations, but a substantial proportion of users are failing to transition into active content creators. Low activation rates can negatively affect long-term engagement and platform growth.

Recommendation

Implement onboarding campaigns, posting tutorials, and first-post incentives to encourage content creation among inactive users.

3.3 Content Performance Analysis

Finding

Photo ID 145 uploaded by Zack_Kemmer93 received 48 likes, making it the most engaging piece of content on the platform.

Business Implication

The existence of highly successful content creators demonstrates opportunities to better understand the characteristics that drive user engagement. Insights from top-performing posts can inform content recommendations and creator support strategies.

Recommendation

Analyse engagement patterns of top-performing content and promote similar content formats through recommendation systems and creator-focused initiatives.

3.4 Hashtag Effectiveness

Finding

The most frequently used hashtags were smile (59), beach (42), party (39), fun (38), and concert (24).

Business Implication

User-generated content is strongly concentrated around lifestyle, entertainment, and positive social experiences. These themes appear to resonate most strongly with the platform community.

Recommendation

Align marketing campaigns and partnership strategies with trending content themes to maximise visibility and audience engagement.

3.5 User Acquisition Trends

Finding

Thursday and Sunday recorded the highest number of user registrations, with 16 new users registering on each day.

Business Implication

Although registrations are relatively evenly distributed across the week, Thursday and Sunday demonstrate slightly stronger acquisition performance and may represent more favourable periods for marketing activity.

Recommendation

Schedule acquisition campaigns, promotional activities, and advertising initiatives around peak registration days to maximise conversion potential.

4. Platform Engagement and Investor-Oriented Metrics

4.1 User Engagement

The platform contains 257 photos generated by 100 users, resulting in an average of 2.57 posts per user.

However, average values conceal considerable variation. Distribution analysis revealed that most users contribute only a small number of posts, while a minority of highly active users generate a disproportionate share of platform content.

This concentration of activity is common within social media ecosystems but presents long-term retention risks if content creation remains dependent on a limited number of users.

4.2 Bot and Automated Behaviour Detection

Thirteen users were identified as having liked every photo available on the platform.

Given the volume of content available, such behaviour is unlikely to occur naturally and may indicate automated engagement activity.

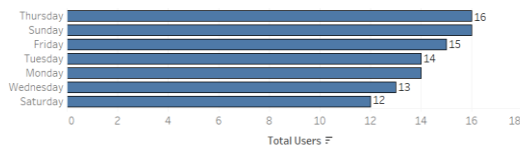
While additional investigation would be required before definitive classification, these accounts should be prioritised for behavioural review.

5. Dashboard and Visual Analytics

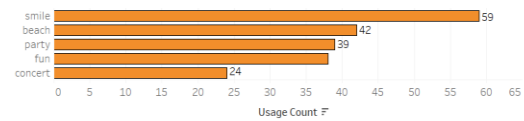
Instagram User Analytics Dashboard

Users: 100 | Photos: 257 | Likes: 8,782 | Avg Posts/User: 2.57 | Inactive Users: 26

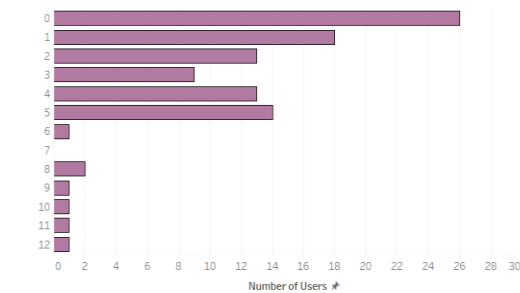
User Registrations by Weekday



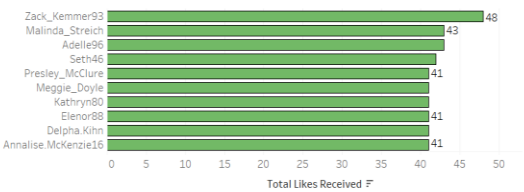
Top Hashtags



User Posting Activity



Top 10 Most Liked Photos



The dashboard consolidates four key analytical perspectives:

- User registration trends
- Hashtag popularity
- User posting behaviour
- Top-performing content

The dashboard highlights four dimensions of platform performance. Registration analysis identifies peak acquisition periods, hashtag analysis reveals content trends, posting activity visualises engagement distribution across users, and photo performance analysis highlights highly engaging content. Together, these visualisations support data-driven marketing and product decisions.

Together, these visualisations provide a concise overview of platform engagement and growth dynamics.

6. Strategic Recommendations

1. Develop targeted activation campaigns for users who have never posted content
2. Reward highly engaged and long-tenured users through loyalty initiatives
3. Schedule acquisition campaigns around Thursday and Sunday registration peaks
4. Incorporate high-performing hashtags into marketing activities
5. Establish monitoring mechanisms for suspicious engagement patterns

7. Limitations and Future Work

The analysis was conducted on a sample dataset and therefore may not fully represent real-world Instagram user behaviour.

Future research could incorporate:

- Time-series engagement analysis
- User segmentation and clustering
- Predictive modelling for user retention
- Content recommendation analytics
- Social network analysis using follower relationships

8. Business Impact

Business Impact: The analysis generated insights across three key areas.

User Growth: Registration activity peaked on Thursday and Sunday, providing evidence for more targeted campaign scheduling.

User Engagement: Twenty-six percent of users had never uploaded content, highlighting opportunities to improve activation and retention.

Platform Integrity: Thirteen accounts exhibited highly unusual engagement behaviour by liking every photo on the platform, indicating potential automated activity.

Collectively, these findings demonstrate how data-driven analysis can support marketing optimisation, engagement improvement, and platform monitoring decisions.

9. Conclusion

This project demonstrates how SQL and Tableau can be combined to transform raw relational data into actionable business intelligence. The findings highlight both growth opportunities and platform risks, providing evidence-based recommendations that support marketing strategy, user engagement, and operational decision-making.

10. Appendix A

Query A1: Loyal User Identification

The screenshot shows a SQL editor window with the following query:

```

1 SELECT
2     id,
3     username,
4     created_at
5 FROM users
6 ORDER BY created_at ASC
7 LIMIT 5;

```

Below the editor is the "Result Grid" showing the results of the query. The grid has three columns: id, username, and created_at. The results are as follows:

	id	username	created_at
▶	80	Darby_Herzog	2016-05-06 00:14:21
	67	Emilio_Bernier52	2016-05-06 13:04:30
	63	Elenor88	2016-05-08 01:30:41
	95	Nicole71	2016-05-09 17:30:22
	38	Jordyn.Jacobson2	2016-05-14 07:56:26
*	NULL	NULL	NULL

Query A2: Inactive User Detection

```
1 SELECT
2     u.id,
3     u.username
4 FROM users u
5 LEFT JOIN photos p
6     ON u.id = p.user_id
7 WHERE p.id IS NULL;
```

Result Grid | | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

	id	username
▶	5	Aniya_Hackett
	7	Kasandra_Homenick
	14	Jadyn81
	21	Rocio33
	24	Maxwell.Halvorson
	25	Tierra.Trantow
	34	Pearl7
	36	Ollie_Ledner37
	41	Mckenna17
	45	David.Osinski47

Query A3: Contest Winner Analysis

```
1 • SELECT
2     u.id,
3     u.username,
4     p.id AS photo_id,
5     COUNT(l.photo_id) AS total_likes
6 FROM photos p
7 JOIN users u
8     ON p.user_id = u.id
9 JOIN likes l
10    ON p.id = l.photo_id
11 GROUP BY p.id, u.id, u.username
12 ORDER BY total_likes DESC
13 LIMIT 1;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:	Fetch rows:
id	username	photo_id	total_likes	
52	Zack_Kemmer93	145	48	

Query A4: Hashtag Popularity Analysis

```
1 • SELECT
2     t.tag_name,
3     COUNT(pt.tag_id) AS usage_count
4 FROM tags t
5 JOIN photo_tags pt
6     ON t.id = pt.tag_id
7 GROUP BY t.id, t.tag_name
8 ORDER BY usage_count DESC
9 LIMIT 5;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:	Fetch rows:
tag_name	usage_count			
smile	59			
beach	42			
party	39			
fun	38			
concert	24			

Query A5: User Registration Trend Analysis

```
1 • SELECT
2     DAYNAME(created_at) AS registration_day,
3     COUNT(*) AS total_users
4 FROM users
5 GROUP BY registration_day
6 ORDER BY total_users DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	registration_day	total_users
▶	Thursday	16
	Sunday	16
	Friday	15
	Tuesday	14
	Monday	14
	Wednesday	13
	Saturday	12

Query A6: User Engagement Assessment

```
1 • SELECT
2     (SELECT COUNT(*) FROM photos) AS total_photos,
3     (SELECT COUNT(*) FROM users) AS total_users,
4     ROUND(
5         (SELECT COUNT(*) FROM photos) /
6         (SELECT COUNT(*) FROM users),
7         2
8     ) AS avg_posts_per_user;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	total_photos	total_users	avg_posts_per_user
▶	257	100	2.57

Query A7: Potential Bot Detection

1

•

SELECT

2

u.id,

3

u.username,

4

COUNT(l.photo_id) AS total_likes

5

FROM users u

6

JOIN likes l

7

ON u.id = l.user_id

8

GROUP BY u.id, u.username

9

HAVING COUNT(l.photo_id) = (

10

SELECT COUNT(*)

11

FROM photos

12

);

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	id	username	total_likes
▶	5	Aniya_Hackett	257
	14	Jadyn81	257
	21	Rocio33	257
	24	Maxwell.Halvorson	257
	36	Ollie_Ledner37	257
	41	Mckenna17	257
	54	Duane60	257
	57	Julien_Schmidt	257
	66	Mike.Auer39	257
	71	Nia_Haag	257
	75	Leslie67	257
	76	Janelle.Nikolaus81	257
	91	Bethany20	257